

Similar Pairs of Strings

Problem Description

You are given two arrays of strings, `s` and `t`, each of length `n`. Each pair `(s[i], t[i])` contains two lowercase English strings.

Two strings are considered **similar** if:

- For every letter x (from 'a' to 'z'), the absolute difference in how many times x appears in the two strings is at most 3.

Your Task:

- For each pair `(s[i], t[i])`, check whether the strings are similar.
- Return an array of `n` elements, where each element is "YES" if the pair is similar, or "NO" otherwise.

Example

Suppose `s = ["aabaab", "aaaaabb"]` and `t = ["bbabbc", "abbbbbbb"]`

Output: `["YES", "NO"]`

Analysis of the first pair

Letter	s[0] count	t[0] count	Difference
a	4	1	3
b	2	4	2
c	0	1	1

The differences for 'a', 'b', and 'c' never exceed 3. This pair is **similar**.

Analysis of the second pair

Letter	s[1] count	t[1] count	Difference
a	5	1	4
b	2	6	4

The differences for 'a' and 'b' violate the condition ($4 > 3$). This pair is **not similar**.

Constraints

- $1 \leq n \leq 5$
- $1 \leq \text{length of any string in the input} \leq 10^5$

Java Implementation

```
import java.util.*;

class Result {

    /*
     * Complete the 'areSimilar' function below.
     *
     * The function is expected to return a STRING_ARRAY.
     * The function accepts following parameters:
     * 1. STRING_ARRAY s
     * 2. STRING_ARRAY t
     */

    public static List<String> areSimilar(List<String> s, List<String> t) {
        List<String> result = new ArrayList<>();
        int n = s.size();

        for (int i = 0; i < n; i++) {
            if (checkSimilarity(s.get(i), t.get(i))) {
                result.add("YES");
            } else {
                result.add("NO");
            }
        }
        return result;
    }
}
```

```
private static boolean checkSimilarity(String s1, String s2) {  
    int[] countS1 = new int[26];  
    int[] countS2 = new int[26];  
  
    // Count frequencies for string s1  
    for (char c : s1.toCharArray()) {  
        countS1[c - 'a']++;  
    }  
  
    // Count frequencies for string s2  
    for (char c : s2.toCharArray()) {  
        countS2[c - 'a']++;  
    }  
  
    // Compare absolute differences  
    for (int i = 0; i < 26; i++) {  
        if (Math.abs(countS1[i] - countS2[i]) > 3) {  
            return false;  
        }  
    }  
    return true;  
}
```